机器学习

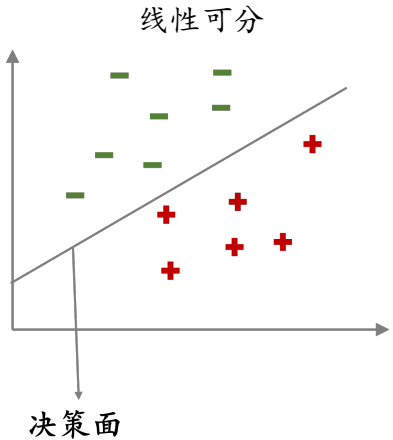
第 5 章 神经网络

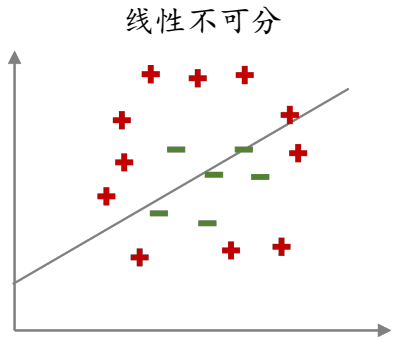
詹大为

计算机与人工智能学院

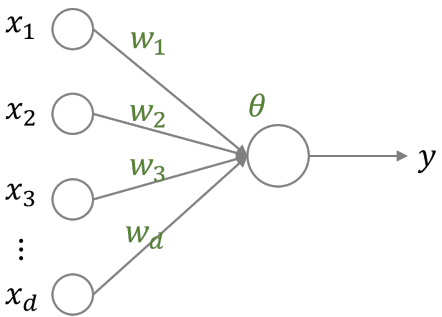
zhandawei@swjtu.edu.cn

1. 感知机模型：线性可分与线性不可分



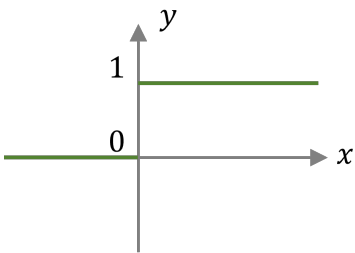


1. 感知机模型

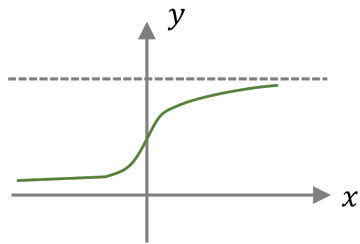


$$
\begin{aligned}
y &= f(w_1x_1 + w_2x_2 + \cdots + w_dx_d - \theta) \\
&= f\left(\sum_{i=1}^{d} w_ix_i - \theta\right) \\
&= f\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} - \theta\right)
\end{aligned}
$$

1. 感知机模型：激活函数



sgn阶跃函数

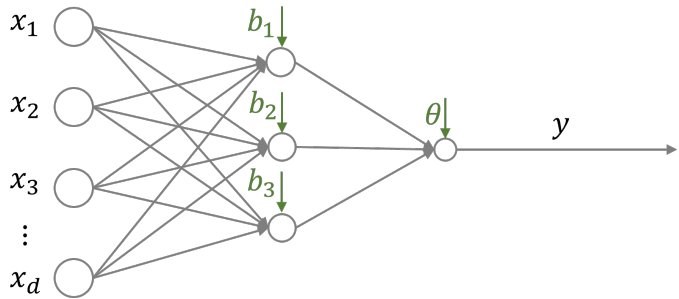


sigmoid函数

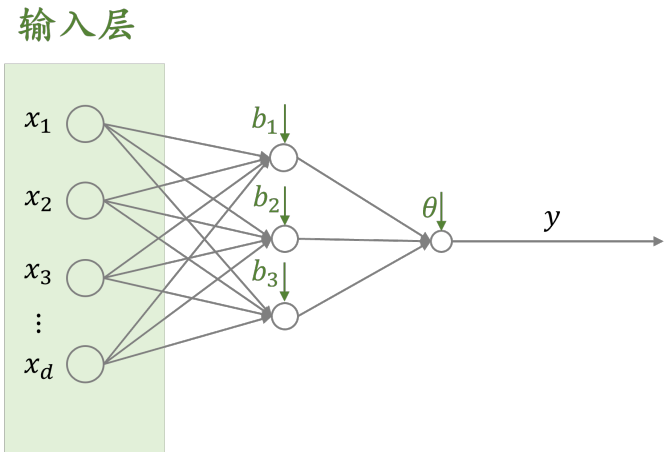
$$y(\boldsymbol{x}) = \text{sgn}\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} - \theta\right)$$

$$y(\boldsymbol{x}) = \text{sigmoid}\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} - \theta\right) = \frac{1}{1 + e^{(-\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + \theta)}}$$

2. 神经网络：基本结构

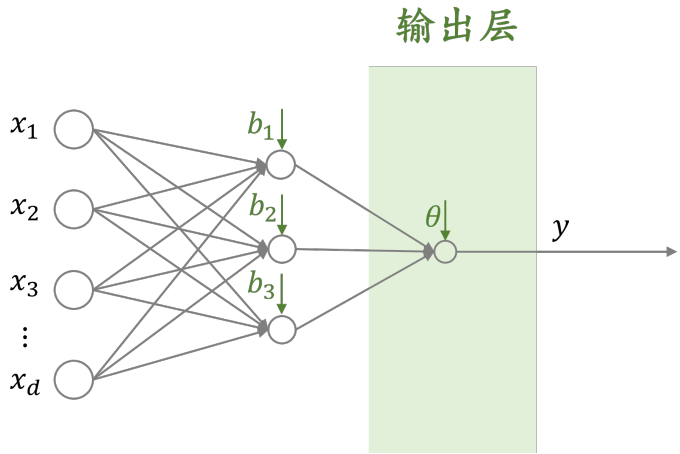


2. 神经网络：基本结构



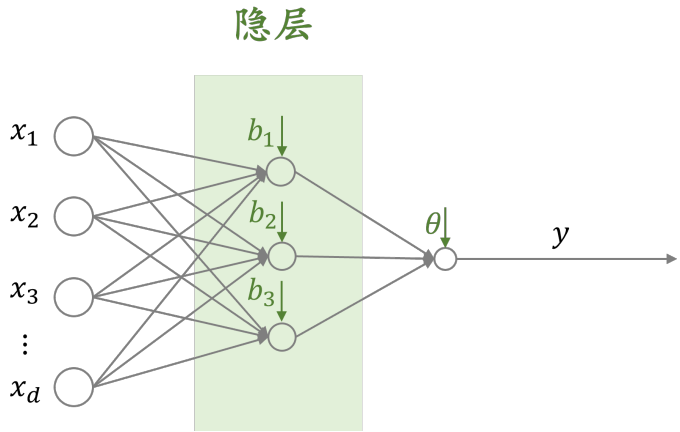
- 输入层没有神经元
- 节点数与特征数目相同

2. 神经网络：基本结构



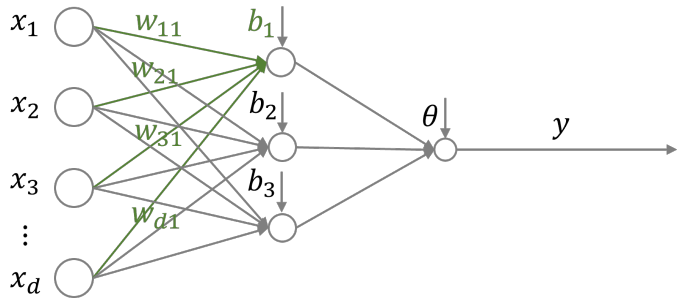
- 输出层有神经元
- 回归问题 1 个节点，k 分类问题 k 个节点

2. 神经网络：基本结构

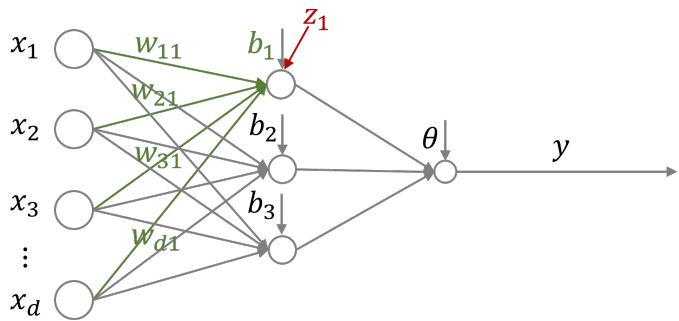


- 隐含层有神经元
- 隐含层层数，每层节点数可以看作超参数

2. 神经网络：前向传播



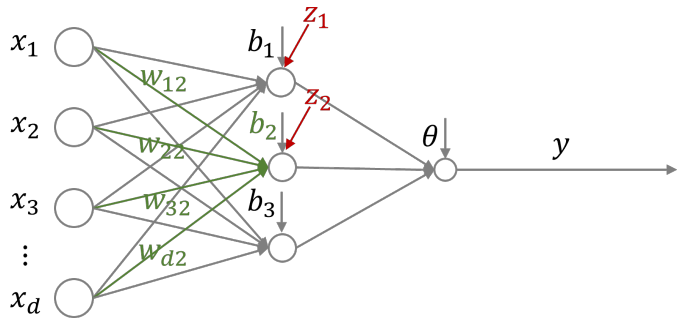
2. 神经网络：前向传播



$$z_1 = w_{11}x_1 + w_{21}x_2 + \cdots + w_{d1}x_d - b_1$$

$$z_1 = \boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{x} - b_1, \quad \boldsymbol{w}_1 = [w_{11}, w_{21}, \cdots, w_{d1}]^{\mathrm{T}}$$

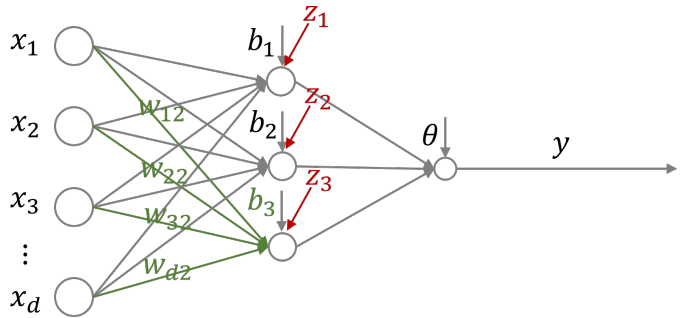
2. 神经网络：前向传播



$$z_1 = \boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{x} - b_1, \quad \boldsymbol{w}_1 = [w_{11}, w_{21}, \cdots, w_{d1}]^{\mathrm{T}}$$

$$z_2 = \boldsymbol{w}_2^{\mathrm{T}}\boldsymbol{x} - b_2, \quad \boldsymbol{w}_2 = [w_{12}, w_{22}, \cdots, w_{d2}]^{\mathrm{T}}$$

2. 神经网络：前向传播

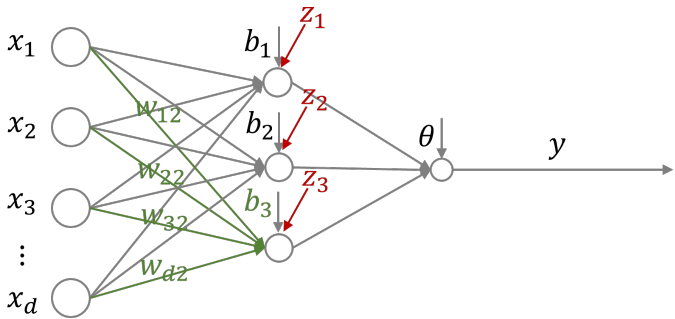


$$z_1 = \boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{x} - b_1, \quad \boldsymbol{w}_1 = [w_{11}, w_{21}, \cdots, w_{d1}]^{\mathrm{T}}$$

$$z_2 = \boldsymbol{w}_2^{\mathrm{T}}\boldsymbol{x} - b_2, \quad \boldsymbol{w}_2 = [w_{12}, w_{22}, \cdots, w_{d2}]^{\mathrm{T}}$$

$$z_3 = \boldsymbol{w}_3^{\mathrm{T}}\boldsymbol{x} - b_3, \quad \boldsymbol{w}_3 = [w_{13}, w_{23}, \cdots, w_{d3}]^{\mathrm{T}}$$

2. 神经网络：前向传播



$$z_1 = \boldsymbol{w}_1^{\mathrm{T}}\boldsymbol{x} - b_1$$

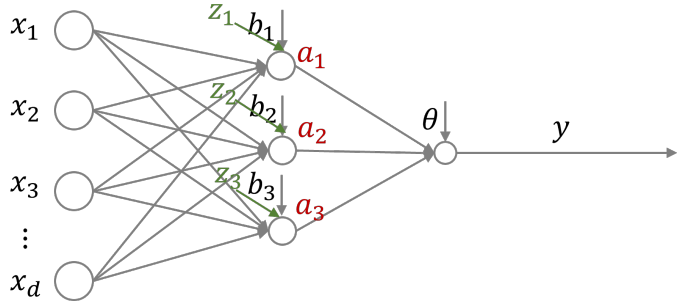
$$z_2 = \boldsymbol{w}_2^{\mathrm{T}}\boldsymbol{x} - b_2$$

$$z_3 = \boldsymbol{w}_3^{\mathrm{T}}\boldsymbol{x} - b_3$$

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \quad \boldsymbol{W} = \begin{bmatrix} \boldsymbol{w}_1 & \boldsymbol{w}_2 & \boldsymbol{w}_3 \end{bmatrix} = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{32} \\ \vdots & \vdots & \vdots \\ w_{d1} & w_{d2} & w_{d3} \end{bmatrix} \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \quad \boldsymbol{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix}$$

$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b}$$

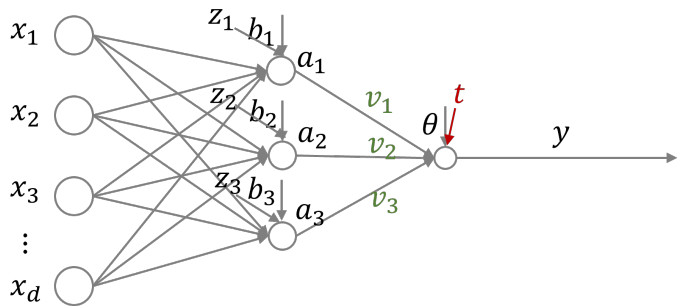
2. 神经网络：前向传播



$$a_1 = \sigma(z_1)$$
$$a_2 = \sigma(z_2)$$
$$a_3 = \sigma(z_3)$$

$$\boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z})$$

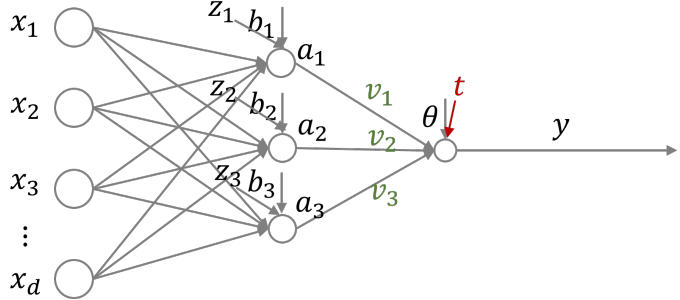
2. 神经网络：前向传播



$$t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta$$

$$y = \sigma(t)$$

2. 神经网络：前向传播



$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b}$$

$$\boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z})$$

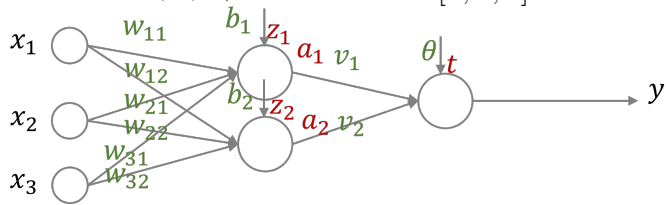
$$t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta$$

$$y = \sigma(t)$$

$$y = \sigma\left(\boldsymbol{v}^{\mathrm{T}}\boldsymbol{\sigma}\left(\boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b}\right) - \theta\right)$$

2. 神经网络：作业 5

有如图所示的神经网络结构，网络所有节点激活函数均为 sigmoid 函数，其中 $w_{11}=2$，$w_{21}=1$，$w_{31}=4$，$w_{12}=-2$，$w_{22}=-3$，$w_{32}=1$，$b_1=14$，$b_2=-4$，$v_1=1$，$v_2=2$，$\theta=2$。试计算神经网络对输入 $[1,2,3]^{\mathrm{T}}$ 的输出值。



$$z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1 = 2\times 1 + 1\times 2 + 4\times 3 - 14 = 2$$

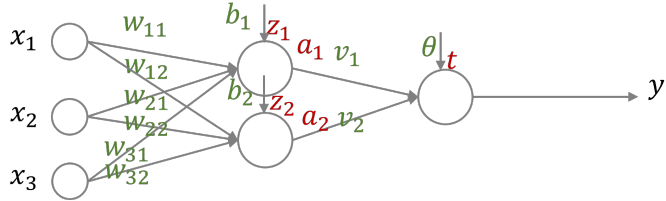
$$z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2 = -2\times 1 - 3\times 2 + 1\times 3 - (-4) = -1$$

$$a_1 = \sigma(z_1) = 0.8808, a_2 = \sigma(z_2) = 0.2689$$

$$t = v_1a_1 + v_2a_2 - \theta = 1\times 0.8808 + 2\times 0.2689 - 2 = -0.5814$$

$$y = \sigma(t) = 0.3586$$

3. 神经网络的训练：误差逆传播



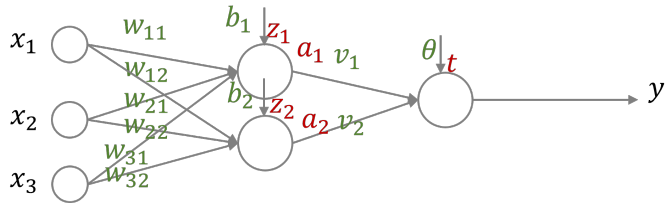
$$z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1 \quad a_1 = \sigma(z_1)$$
$$z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2 \quad a_2 = \sigma(z_2)$$

$$t = v_1a_1 + v_2a_2 - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$L(w_{11}, w_{21}, w_{31}, w_{12}, w_{22}, w_{32}, b_1, b_2, v_1, v_2, \theta) =$$
$$\frac{1}{2}\left[\sigma\left(v_1\sigma\left(w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1\right) + v_2\sigma\left(w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2\right) - \theta\right) - h\right]^2$$

$$w_{11}^\star, w_{21}^\star, w_{31}^\star, w_{12}^\star, w_{22}^\star, w_{32}^\star, b_1^\star, b_2^\star, v_1^\star, v_2^\star, \theta^\star =$$
$$\arg\min L(w_{11}, w_{21}, w_{31}, w_{12}, w_{22}, w_{32}, b_1, b_2, v_1, v_2, \theta)$$

3. 神经网络的训练：误差逆传播



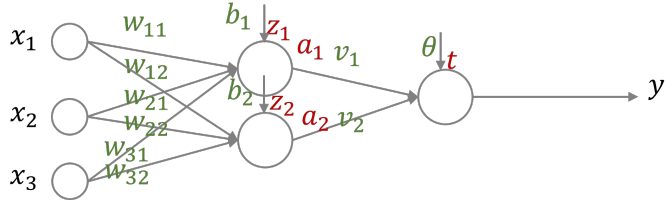
$$w_{11}^\star, w_{21}^\star, w_{31}^\star, w_{12}^\star, w_{22}^\star, w_{32}^\star, b_1^\star, b_2^\star, v_1^\star, v_2^\star, \theta^\star = \arg\min L(w_{11}, w_{21}, w_{31}, w_{12}, w_{22}, w_{32}, b_1, b_2, v_1, v_2, \theta)$$

$$\boldsymbol{\alpha}^\star = \arg\min L(\boldsymbol{\alpha})$$

梯度下降法

$$\boldsymbol{\alpha}_{i+1} = \boldsymbol{\alpha}_i - \eta \nabla L(\boldsymbol{\alpha}_i)$$

3. 神经网络的训练：误差逆传播

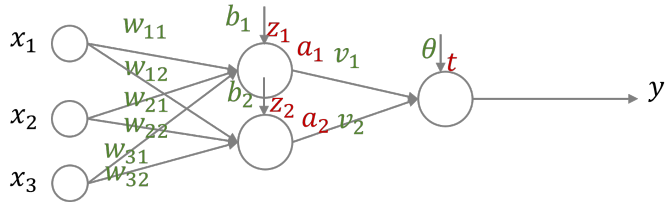


$$z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1 \quad a_1 = \sigma(z_1)$$
$$z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2 \quad a_2 = \sigma(z_2)$$

$$t = v_1a_1 + v_2a_2 - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \theta} = (y-h)y(1-y)(-1)$$

3. 神经网络的训练：误差逆传播



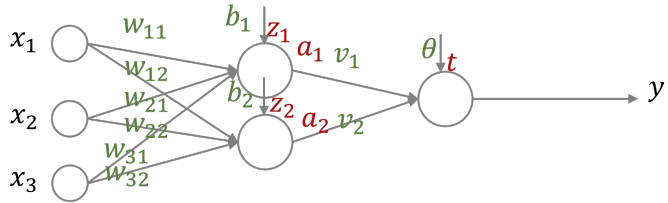
$$z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1 \quad a_1 = \sigma(z_1)$$
$$z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2 \quad a_2 = \sigma(z_2)$$

$$t = v_1a_1 + v_2a_2 - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial v_1} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial v_1} = (y-h)y(1-y)a_1$$

$$\frac{\partial L}{\partial v_2} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial v_2} = (y-h)y(1-y)a_2$$

3. 神经网络的训练：误差逆传播



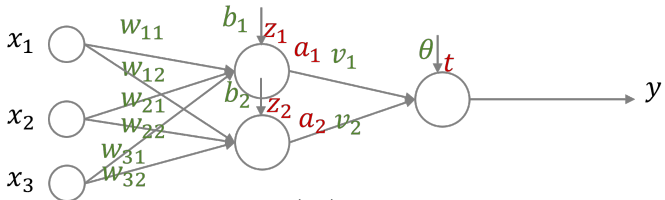
$$z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1 \quad a_1 = \sigma(z_1)$$
$$z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2 \quad a_2 = \sigma(z_2)$$

$$t = v_1a_1 + v_2a_2 - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial b_1} = (y-h)y(1-y)v_1a_1(1-a_1)(-1)$$

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_2}\frac{\partial a_2}{\partial z_2}\frac{\partial z_2}{\partial b_2} = (y-h)y(1-y)v_2a_2(1-a_2)(-1)$$

3. 神经网络的训练：误差逆传播



$z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1 \quad a_1 = \sigma(z_1)$

$z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2 \quad a_2 = \sigma(z_2)$

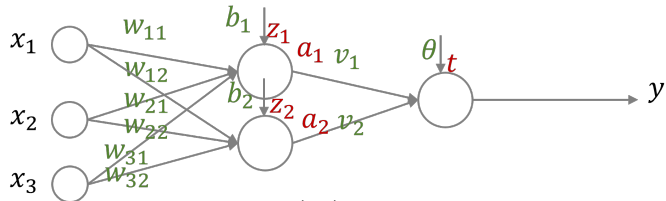
$t = v_1a_1 + v_2a_2 - \theta \quad y = \sigma(t) \quad L = \frac{1}{2}(y-h)^2$

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial w_{11}} = (y-h)y(1-y)v_1a_1(1-a_1)x_1$$

$$\frac{\partial L}{\partial w_{21}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial w_{21}} = (y-h)y(1-y)v_1a_1(1-a_1)x_2$$

$$\frac{\partial L}{\partial w_{31}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial w_{31}} = (y-h)y(1-y)v_1a_1(1-a_1)x_3$$

3. 神经网络的训练：误差逆传播



$$z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1 \quad a_1 = \sigma(z_1)$$

$$z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2 \quad a_2 = \sigma(z_2)$$

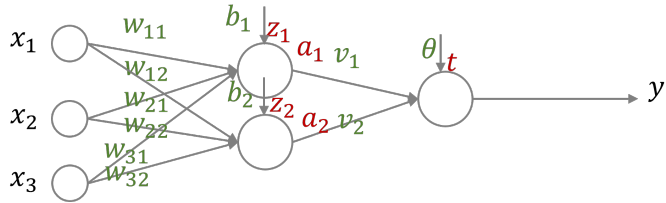
$$t = v_1a_1 + v_2a_2 - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial w_{12}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_2}\frac{\partial a_2}{\partial z_2}\frac{\partial z_2}{\partial w_{12}} = (y-h)y(1-y)v_2a_2(1-a_2)x_1$$

$$\frac{\partial L}{\partial w_{22}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_2}\frac{\partial a_2}{\partial z_2}\frac{\partial z_2}{\partial w_{22}} = (y-h)y(1-y)v_2a_2(1-a_2)x_2$$

$$\frac{\partial L}{\partial w_{32}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_2}\frac{\partial a_2}{\partial z_2}\frac{\partial z_2}{\partial w_{32}} = (y-h)y(1-y)v_2a_2(1-a_2)x_3$$

3. 神经网络的训练：误差逆传播向量形式

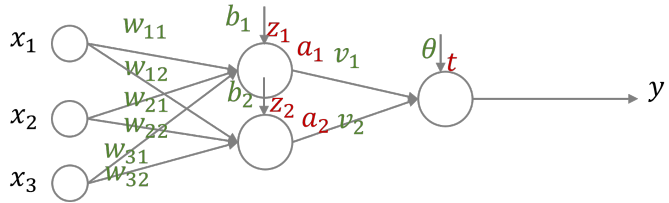


$$z_1 = w_{11}x_1 + w_{21}x_2 + w_{31}x_3 - b_1 \quad a_1 = \sigma(z_1)$$
$$z_2 = w_{12}x_1 + w_{22}x_2 + w_{32}x_3 - b_2 \quad a_2 = \sigma(z_2)$$

$$t = v_1a_1 + v_2a_2 - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b} \qquad \boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z}) \qquad t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

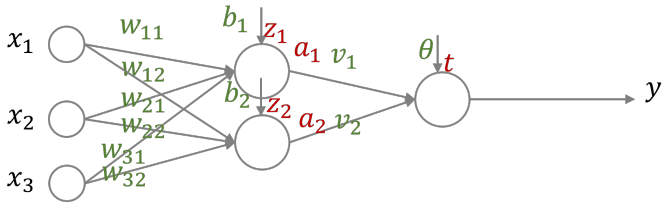
3. 神经网络的训练：误差逆传播向量形式



$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b} \qquad \boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z}) \qquad t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \theta} = (y-h)y(1-y)(-1)$$

3. 神经网络的训练：误差逆传播向量形式

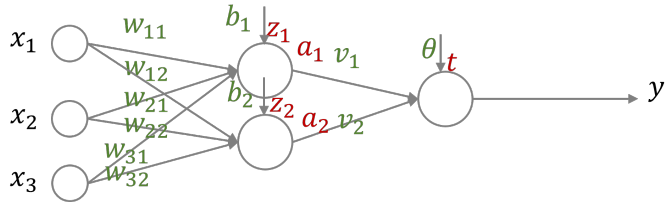


$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b} \qquad \boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z}) \qquad t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial v_1} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial v_1} = (y-h)y(1-y)a_1$$

$$\frac{\partial L}{\partial v_2} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial v_2} = (y-h)y(1-y)a_2$$

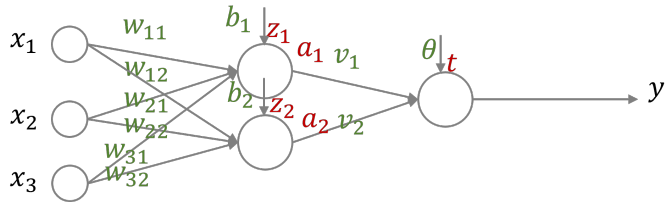
3. 神经网络的训练：误差逆传播向量形式



$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b} \qquad \boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z}) \qquad t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial \boldsymbol{v}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \boldsymbol{v}} = (y-h)y(1-y)\boldsymbol{a}$$

3. 神经网络的训练：误差逆传播向量形式

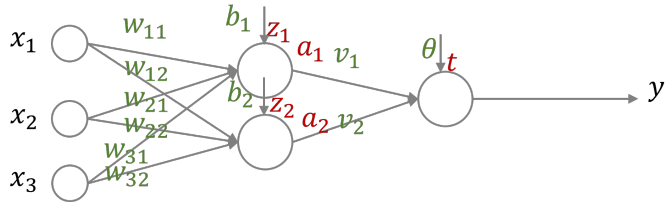


$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b} \qquad \boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z}) \qquad t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta \qquad y = \sigma(t) \quad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_1}\frac{\partial a_1}{\partial z_1}\frac{\partial z_1}{\partial b_1} = (y-h)y(1-y)v_1a_1(1-a_1)(-1)$$

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial a_2}\frac{\partial a_2}{\partial z_2}\frac{\partial z_2}{\partial b_2} = (y-h)y(1-y)v_2a_2(1-a_2)(-1)$$

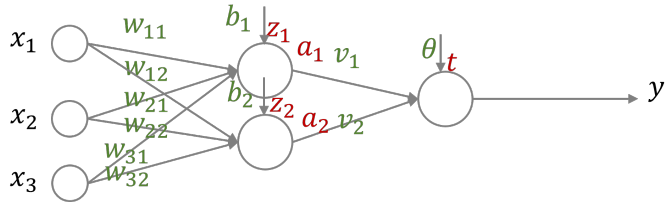
3. 神经网络的训练：误差逆传播向量形式



$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b} \qquad \boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z}) \qquad t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta \qquad y = \sigma(t) \quad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial \boldsymbol{b}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \boldsymbol{a}}\frac{\partial \boldsymbol{a}}{\partial \boldsymbol{z}}\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{b}} = (y-h)y(1-y)\boldsymbol{v} \odot \boldsymbol{a} \odot (1-\boldsymbol{a})(-1)$$

3. 神经网络的训练：误差逆传播向量形式



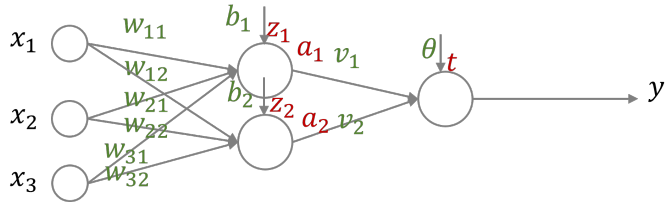
$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b} \qquad \boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z}) \qquad t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial w_{11}} = (y-h)y(1-y)v_1a_1(1-a_1)x_1 \qquad \frac{\partial L}{\partial w_{12}} = (y-h)y(1-y)v_2a_2(1-a_2)x_1$$

$$\frac{\partial L}{\partial w_{21}} = (y-h)y(1-y)v_1a_1(1-a_1)x_2 \qquad \frac{\partial L}{\partial w_{22}} = (y-h)y(1-y)v_2a_2(1-a_2)x_2$$

$$\frac{\partial L}{\partial w_{31}} = (y-h)y(1-y)v_1a_1(1-a_1)x_3 \qquad \frac{\partial L}{\partial w_{32}} = (y-h)y(1-y)v_2a_2(1-a_2)x_3$$

3. 神经网络的训练：误差逆传播向量形式



$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b} \qquad \boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z}) \qquad t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta \qquad y = \sigma(t) \qquad L = \frac{1}{2}(y-h)^2$$

$$\frac{\partial L}{\partial \boldsymbol{W}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \boldsymbol{a}}\frac{\partial \boldsymbol{a}}{\partial \boldsymbol{z}}\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{W}} = \boldsymbol{x}\left[(y-h)y(1-y)\boldsymbol{v} \odot \boldsymbol{a} \odot (1-\boldsymbol{a})\right]^{\mathrm{T}}$$

3. 神经网络的训练：误差逆传播向量形式

$$\boldsymbol{z} = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{x} - \boldsymbol{b} \qquad \boldsymbol{a} = \boldsymbol{\sigma}(\boldsymbol{z}) \qquad t = \boldsymbol{v}^{\mathrm{T}}\boldsymbol{a} - \theta \qquad y = \sigma(t) \quad L = \frac{1}{2}(y-h)^2$$

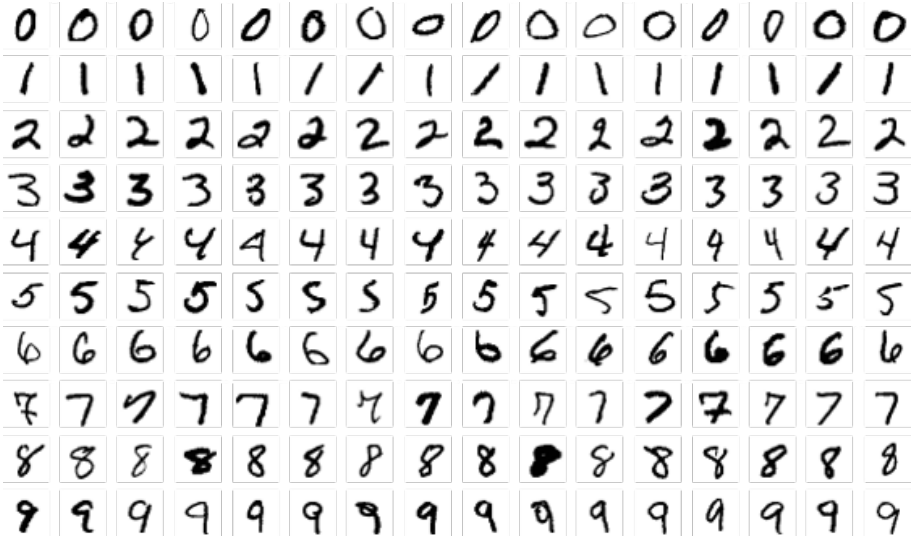
$$\frac{\partial L}{\partial \theta} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \theta} = (y-h)y(1-y)(-1)$$

$$\frac{\partial L}{\partial \boldsymbol{v}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \boldsymbol{v}} = (y-h)y(1-y)\boldsymbol{a}$$

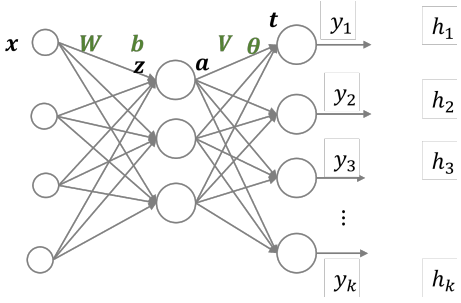
$$\frac{\partial L}{\partial \boldsymbol{b}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \boldsymbol{a}}\frac{\partial \boldsymbol{a}}{\partial \boldsymbol{z}}\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{b}} = (y-h)y(1-y)\boldsymbol{v} \odot \boldsymbol{a} \odot (1-\boldsymbol{a})(-1)$$

$$\frac{\partial L}{\partial \boldsymbol{W}} = \frac{\partial L}{\partial y}\frac{\partial y}{\partial t}\frac{\partial t}{\partial \boldsymbol{a}}\frac{\partial \boldsymbol{a}}{\partial \boldsymbol{z}}\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{W}} = \boldsymbol{x}\left[(y-h)y(1-y)\boldsymbol{v} \odot \boldsymbol{a} \odot (1-\boldsymbol{a})\right]^{\mathrm{T}}$$

3. 神经网络的训练：多分类问题



3. 神经网络的训练：多分类问题



$$\boldsymbol{x}(784\times 1)$$
$$\boldsymbol{W}(784\times 12)$$
$$\boldsymbol{b}(12\times 1)$$
$$\boldsymbol{z}(12\times 1)$$
$$\boldsymbol{a}(12\times 1)$$
$$\boldsymbol{V}(12\times 10)$$
$$\boldsymbol{\theta}(10\times 1)$$
$$\boldsymbol{t}(10\times 1)$$
$$\boldsymbol{y}(10\times 1)$$
$$\boldsymbol{h}(10\times 1)$$

$$\boldsymbol{z}=\boldsymbol{W}^{\mathrm{T}}\boldsymbol{x}-\boldsymbol{b}$$
$$\boldsymbol{a}=\boldsymbol{\sigma}(\boldsymbol{z})$$
$$\boldsymbol{t}=\boldsymbol{V}^{\mathrm{T}}\boldsymbol{a}-\boldsymbol{\theta}$$
$$\boldsymbol{y}=\boldsymbol{\sigma}(\boldsymbol{t})$$
$$L=\frac{1}{2}(\boldsymbol{y}-\boldsymbol{h})^{\mathrm{T}}(\boldsymbol{y}-\boldsymbol{h})$$

4. 实验：多分类问题

$$\boldsymbol{z}=\boldsymbol{W}^{\mathrm{T}}\boldsymbol{x}-\boldsymbol{b},\quad \boldsymbol{a}=\boldsymbol{\sigma}(\boldsymbol{z}),\quad \boldsymbol{t}=\boldsymbol{V}^{\mathrm{T}}\boldsymbol{a}-\boldsymbol{\theta},\quad \boldsymbol{y}=\boldsymbol{\sigma}(\boldsymbol{t}),\quad L=\frac{1}{2}(\boldsymbol{y}-\boldsymbol{h})^{\mathrm{T}}(\boldsymbol{y}-\boldsymbol{h})$$

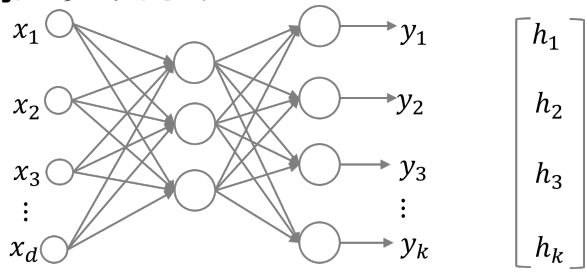
$$\frac{\partial L}{\partial \boldsymbol{\theta}}=\frac{\partial L}{\partial \boldsymbol{y}}\frac{\partial \boldsymbol{y}}{\partial \boldsymbol{t}}\frac{\partial \boldsymbol{t}}{\partial \boldsymbol{\theta}}=(\boldsymbol{y}-\boldsymbol{h})\odot\boldsymbol{y}\odot(1-\boldsymbol{y})(-1)$$

$$\frac{\partial L}{\partial \boldsymbol{V}}=\frac{\partial L}{\partial \boldsymbol{y}}\frac{\partial \boldsymbol{y}}{\partial \boldsymbol{t}}\frac{\partial \boldsymbol{t}}{\partial \boldsymbol{V}}=\boldsymbol{a}\left[(\boldsymbol{y}-\boldsymbol{h})\odot\boldsymbol{y}\odot(1-\boldsymbol{y})\right]^{\mathrm{T}}$$

$$\frac{\partial L}{\partial \boldsymbol{b}}=\frac{\partial L}{\partial \boldsymbol{y}}\frac{\partial \boldsymbol{y}}{\partial \boldsymbol{t}}\frac{\partial \boldsymbol{t}}{\partial \boldsymbol{a}}\frac{\partial \boldsymbol{a}}{\partial \boldsymbol{z}}\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{b}}=\boldsymbol{V}\left[(\boldsymbol{y}-\boldsymbol{h})\odot\boldsymbol{y}\odot(1-\boldsymbol{y})\right]\odot\boldsymbol{a}\odot(1-\boldsymbol{a})(-1)$$

$$\frac{\partial L}{\partial \boldsymbol{W}}=\frac{\partial L}{\partial \boldsymbol{y}}\frac{\partial \boldsymbol{y}}{\partial \boldsymbol{t}}\frac{\partial \boldsymbol{t}}{\partial \boldsymbol{a}}\frac{\partial \boldsymbol{a}}{\partial \boldsymbol{z}}\frac{\partial \boldsymbol{z}}{\partial \boldsymbol{W}}=\boldsymbol{x}\left\{\boldsymbol{V}\left[(\boldsymbol{y}-\boldsymbol{h})\odot\boldsymbol{y}\odot(1-\boldsymbol{y})\right]\odot\boldsymbol{a}\odot(1-\boldsymbol{a})\right\}^{\mathrm{T}}$$

3. 神经网络的训练：多分类问题



一个样本损失

$$L = \frac{1}{2}(\boldsymbol{y} - \boldsymbol{h})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{h})$$

所有样本总损失

$$L_{\text{total}} = \frac{1}{2}\sum_{j=1}^{m}(\boldsymbol{y}_j - \boldsymbol{h}_j)^{\mathrm{T}}(\boldsymbol{y}_j - \boldsymbol{h}_j)$$

3. 神经网络的训练：多分类问题

一个样本的损失，**随机梯度下降**

$$L = \frac{1}{2}(\boldsymbol{y} - \boldsymbol{h})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{h})$$

$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i - \eta \frac{\partial L}{\partial \boldsymbol{\theta}_i}$$

$$\boldsymbol{V}_{i+1} = \boldsymbol{V}_i - \eta \frac{\partial L}{\partial \boldsymbol{V}_i}$$

$$\boldsymbol{b}_{i+1} = \boldsymbol{b}_i - \eta \frac{\partial L}{\partial \boldsymbol{b}_i}$$

$$\boldsymbol{W}_{i+1} = \boldsymbol{W}_i - \eta \frac{\partial L}{\partial \boldsymbol{W}_i}$$

m 个样本的总损失，**标准梯度下降**

$$L_{\text{total}} = \frac{1}{2}\sum_{j=1}^{m}(\boldsymbol{y}_j - \boldsymbol{h}_j)^{\mathrm{T}}(\boldsymbol{y}_j - \boldsymbol{h}_j)$$

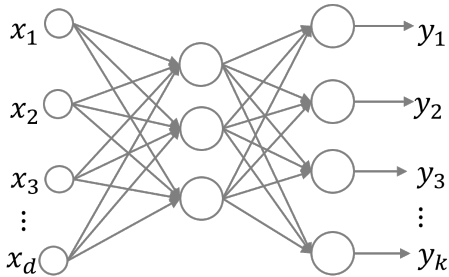
$$\boldsymbol{\theta}_{i+1} = \boldsymbol{\theta}_i - \eta \sum_{j=1}^{m} \frac{\partial L_j}{\partial \boldsymbol{\theta}_i}$$

$$\boldsymbol{V}_{i+1} = \boldsymbol{V}_i - \eta \sum_{j=1}^{m} \frac{\partial L_j}{\partial \boldsymbol{V}_i}$$

$$\boldsymbol{b}_{i+1} = \boldsymbol{b}_i - \eta \sum_{j=1}^{m} \frac{\partial L_j}{\partial \boldsymbol{b}_i}$$

$$\boldsymbol{W}_{i+1} = \boldsymbol{W}_i - \eta \sum_{j=1}^{m} \frac{\partial L_j}{\partial \boldsymbol{W}_i}$$

3. 神经网络的训练：多分类问题损失函数



$$\begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_k \end{bmatrix}$$

平方损失

$$L = \frac{1}{2}(\boldsymbol{y} - \boldsymbol{h})^{\mathrm{T}}(\boldsymbol{y} - \boldsymbol{h})$$
$$= \frac{1}{2}\sum_{i=1}^{k}(y_i - h_i)^2$$

交叉熵损失

$$L = -\sum_{i=1}^{k} y_i \log_2(h_i)$$